

Vidhi Mistry

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EDUCATION

University of Illinois Urbana-Champaign

Master of Engineering in Chemical Engineering Leadership | GPA: 3.86

Illinois, USA

Aug 2025 – Present

Nirma University

Bachelor of Technology in Chemical Engineering | GPA: 3.24

Ahmedabad, Gujarat, India

Oct 2021 – May 2025

EXPERIENCE

University of Illinois Urbana-Champaign

Graduate Research Assistant

Urbana-Champaign, IL

May 2026 – Present

- Assisted in developing machine learning models to identify safer, biodegradable alternatives to PFAS by predicting molecular toxicity, reducing the experimental effort required for chemical screening.
- Worked on a team building an end-to-end molecular machine learning workflow in PyTorch, covering data preparation, model training, cross-validation and testing on unseen data.
- Applied MolFormer, a pretrained transformer model, to generate molecular representations directly from chemical structures.

University of Illinois Urbana-Champaign

Graduate Research Assistant, RIPE Project

Urbana-Champaign, IL

Jun 2026 – Present

- Authored 8+ Standard Operating Procedures and safety documentation covering autoclave operation, equipment sterilization and experimental procedures, standardizing laboratory practice across the group.
- Prepared chemical solutions for sterilizing laboratory and greenhouse equipment and carried out autoclave sterilization for a 192-plant soybean study across multiple treatment groups.
- Analyzed soybean phenotyping datasets in R to evaluate Water Use Efficiency and prepared results for research reporting.
- Worked in a cleanroom environment, following contamination-control and laboratory safety protocols throughout experimental work.

Gujarat Narmada Valley Fertilizers & Chemicals Ltd

Process Engineering Trainee

Bharuch, Gujarat, India

May 2024 – Jun 2024

- Documented Standard Operating Procedures for plant shutdown, equipment isolation, depressurization, purging and lockout-tagout, and observed a scheduled maintenance shutdown of an operating plant.
- Monitored operating parameters including temperature, pressure and flow using Distributed Control Systems under engineering supervision.
- Applied learning-from-incident findings when documenting Standard Operating Procedures, with exposure to HAZOP preparation in an operating plant.

Nirma University

Researcher

Ahmedabad, Gujarat, India

Dec 2024 – May 2025

"Employment of Industrial Waste Material: Calcium Carbide Residue (CCR) as a Photocatalyst for Green Hydrogen Production".

- Filed a patent application (pending) for a novel method using industrial waste Calcium Carbide Residue as a photocatalyst, achieving approximately 15% higher hydrogen generation efficiency than electrolysis alone by combining electrolysis with photocatalysis.
- Produced hydrogen at an applied potential of 1.3 V, below the 1.6-2.2 V practical operating range of PEM and alkaline electrolyzers, and quantified hydrogen yield by water displacement collected at the cathode.
- Deposited CCR catalyst onto electrodes by sol-gel method and applied a polyvinyl alcohol overcoat to prevent catalyst detachment during operation.
- Designed a custom photo-electrocatalytic hydrogen production cell in AutoCAD and had it externally fabricated for experimental use.

Nirma University

Researcher

Ahmedabad, Gujarat, India

Jul 2023 – Sep 2023

"Wastewater Treatment of Dye Intermediate Industry for Zero-Liquid Discharge".

- Performed 15+ lab-scale experiments evaluating coagulant and flocculant combinations for dye-intermediate industry wastewater.
- Measured Chemical Oxygen Demand by the closed reflux colorimetric method and Biological Oxygen Demand by the dilution method.
- Reduced Chemical Oxygen Demand from the thousands of mg/L to approximately 100 mg/L and Biological Oxygen Demand to approximately 30 mg/L through optimized chemical dosing.
- Developed and applied modified chitosan as a biodegradable flocculant.

RESEARCH & PROJECTS

University of Illinois Urbana-Champaign

Graduate Research Project

Urbana-Champaign, IL

Feb 2026 – May 2026

"AI-Assisted Physics-Guided Modelling for Carbon Capture and Gas Separation in Polymer Membranes"

- Developed predictive machine learning models, Random Forest as baseline and XGBoost as the main model, combining Morgan fingerprints with physics-based gas transport and polymer descriptors to evaluate membrane selectivity across three gas pairs for carbon capture.
- Implemented GroupShuffleSplit and 5-fold GroupKFold validation grouped by polymer SMILES to ensure no polymer appearing in training was present in the test set.
- Achieved R^2 of 0.828 and RMSE of 0.285 on a test set of polymers unseen by the model, with a 5-fold cross-validated R^2 of 0.665 ± 0.157 .
- Identified physics-based and polymer descriptors as the strongest predictors of selectivity in feature importance analysis, ranking above structural fingerprints, with higher-rigidity polymer structures associated with higher selectivity.

TECHNICAL SKILLS

Laboratory & Materials: Sol-gel catalyst deposition, Electrode fabrication and coating, PVA binder overcoating, Photo-electrocatalytic hydrogen generation, COD and BOD analysis, Coagulation-flocculation, Autoclave sterilization, Contamination control

Computational & Methods: Python, R, PyTorch, XGBoost, scikit-learn, RDKit, PyMOL, Molecular dynamics simulation, AutoCAD, Leakage-controlled model validation, Experimental design, SOP development, Six Sigma Green Belt (LinkedIn Learning)